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Department of Electrical and Electronic Engineering

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Experiment 05

Verification of Superposition Theorem and Maximum Power Transfer Theorem for a DC Network

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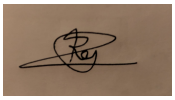
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Verification of Superposition Theorem and Maximum Power Transfer Theorem for a DC Network"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit (EEE 0713 1206B)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

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Objective

- To examine the Superposition Theorem in multi-source linear DC networks.
- To experimentally prove that total branch currents equal the algebraic sum of responses from individual independent sources.
- To investigate the Maximum Power Transfer Theorem by observing load power variations across variable resistances.
- To trace the power-resistance profile experimentally and compare peak power conditions with theoretical predictions.

Introduction

This laboratory study evaluates two fundamental principles governing linear DC networks. The **Superposition Theorem** simplifies multi-source network analysis by calculating the isolated contribution of each independent source while deactivating others. The **Maximum Power Transfer Theorem** identifies the specific load condition that extracts peak power from a linear network. Both principles provide indispensable tools for network reduction and power optimization in resistive circuits.

Theory

Part A: Superposition Theorem

Statement of the Theorem

> *In any linear, bilateral DC network containing multiple independent sources, the response in any branch equals the algebraic sum of the individual responses produced by each source acting independently while all other independent sources are deactivated.*

Deactivation of Independent Sources

- **Ideal Voltage Sources** are set to zero by replacing them with a **short circuit**.
- **Ideal Current Sources** are set to zero by replacing them with an **open circuit**.

Mathematical Formulation

For a branch current I driven by sources V_1 and V_2:

$$I = I' + I''$$

where I' is the branch current with V_1 active (V_2 shorted), and I'' is the branch current with V_2 active (V_1 shorted).

Non-Applicability to Power Calculations

Because electrical power follows a non-linear relationship ($P = I^2R$), superposition cannot be applied directly to power:

$$P_{\text{total}} \neq P_1 + P_2.$$

Total current or voltage must be determined prior to calculating network power.

Part B: Maximum Power Transfer Theorem

Statement of the Theorem

A linear DC source network transfers maximum power to an external load R_L when the load resistance matches the internal Thevenin resistance R_{th} of the network:

$$R_L = R_{th}.$$

Derivation

For a network represented by V_{th} and R_{th} driving R_L:

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad P_L = I_L^2 R_L = \frac{V_{th}^2 R_L}{(R_{th} + R_L)^2}.$$

Setting $\dfrac{dP_L}{dR_L} = 0$ yields $R_L = R_{th}$, resulting in:

$$I_{L(\max)} = \frac{V_{th}}{2R_{th}}, \qquad P_{L(\max)} = \frac{V_{th}^2}{4R_{th}}.$$

Under maximum power transfer, exactly 50% of the total power is dissipated within the internal source resistance R_{th}.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
Part A: Superposition Theorem		
Dual-Channel DC Power Supply	RPS1, RPS2 (0 V–30 V)	1
Resistor R_1	325 Ω	1
Resistor R_2	550 Ω	1
Resistor R_3	960 Ω	1
Part B: Maximum Power Transfer Theorem		
DC Power Supply (V_{th})	8 V	1
Series Resistor R_i ($= R_{th}$)	663 Ω	1
Decade Resistance Box (R_L)	0 Ω–5 kΩ	1
Common		
Digital Multimeter (DMM)	DC Voltage / Current	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

Circuit Diagrams

Part A: Superposition Theorem

Part B: Maximum Power Transfer Theorem

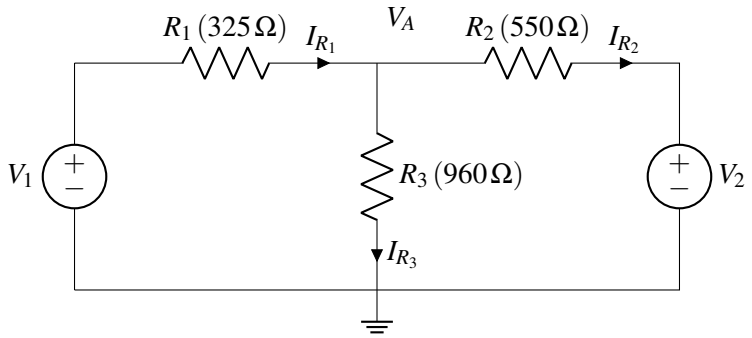


Figure 1: Complete DC network driven simultaneously by voltage sources V_1 and V_2.

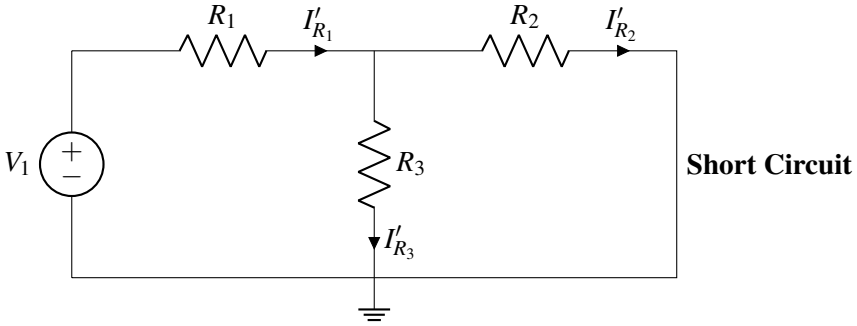


Figure 2: Network response with V_1 active and V_2 short-circuited.

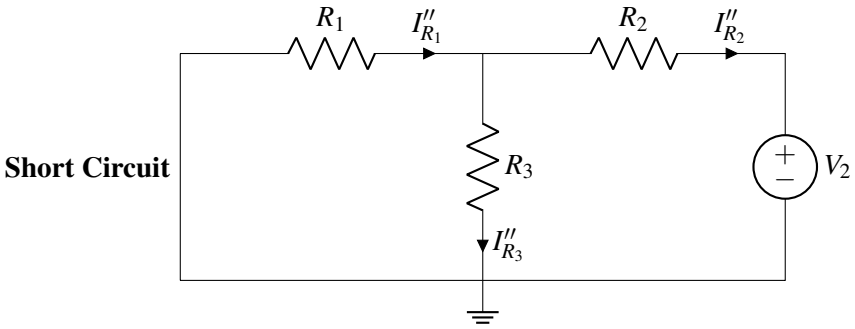


Figure 3: Network response with V_2 active and V_1 short-circuited.

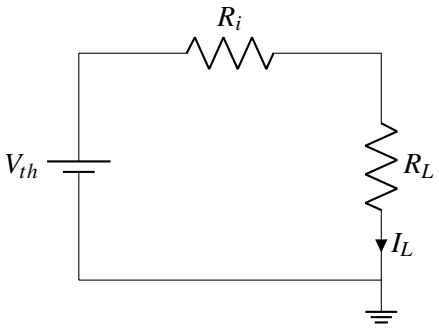


Figure 4: Test circuit for maximum power transfer using $V_{th} = 8\,\text{V}$ and $R_i = 663\,\Omega$.

Experimental Procedure

Part A: Verification of Superposition Theorem

1. Construct the circuit shown in Figure 1 using assigned resistor values ($R_1 = 325\,\Omega$, $R_2 = 550\,\Omega$, $R_3 = 960\,\Omega$).
2. Power both sources (V_1 and V_2) and record the combined current I_{R_3}.
3. Deactivate V_2 using a short circuit (Figure 2) and measure contribution I'_{R_3}.
4. Restore V_2, short-circuit V_1 (Figure 3), and measure contribution I''_{R_3}.
5. Confirm that $I_{R_3} = I'_{R_3} + I''_{R_3}$ and repeat for the second voltage combination.

Part B: Verification of Maximum Power Transfer Theorem

1. Assemble the equivalent network (Figure 4) with $V_{th} = 8\,\text{V}$ and $R_i = 663\,\Omega$.
2. Connect the decade resistance box as variable load R_L.
3. Adjust R_L across values ranging below and above R_i.
4. Measure I_L at each step and evaluate load power P_L.
5. Plot P_L versus R_L to locate the resistance corresponding to maximum power delivery.

Theoretical Calculations

Part A: Superposition Theorem

Using fixed resistances $R_1 = 325\,\Omega$, $R_2 = 550\,\Omega$, $R_3 = 960\,\Omega$:

Case 1: Only V_1 Active (V_2 Short-Circuited)

$$R_{23} = \frac{R_2 R_3}{R_2 + R_3} = 349.67\,\Omega, \qquad R_{\text{eq1}} = R_1 + R_{23} = 674.67\,\Omega.$$

$$I'_{\text{total}} = \frac{V_1}{R_{\text{eq1}}}, \qquad V'_A = I'_{\text{total}} \times R_{23}, \qquad I'_{R_3} = \frac{V'_A}{R_3}.$$

Case 2: Only V_2 Active (V_1 Short-Circuited)

$$R_{13} = \frac{R_1 R_3}{R_1 + R_3} = 242.80\,\Omega, \qquad R_{\text{eq2}} = R_2 + R_{13} = 792.80\,\Omega.$$

$$I''_{\text{total}} = \frac{V_2}{R_{\text{eq2}}}, \qquad V''_A = I''_{\text{total}} \times R_{13}, \qquad I''_{R_3} = \frac{V''_A}{R_3}.$$

Trial 1 ($V_1 = 10\,\text{V}$, $V_2 = 10\,\text{V}$)

$$I'_{R_3} = 5.39\,\text{mA}, \qquad I''_{R_3} = 3.19\,\text{mA}, \qquad I_{R_3} = 8.58\,\text{mA} \approx 8.59\,\text{mA}.$$

Trial 2 ($V_1 = 15\,\text{V}$, $V_2 = 10\,\text{V}$)

$$I'_{R_3} = 8.09\,\text{mA}, \qquad I''_{R_3} = 3.19\,\text{mA}, \qquad I_{R_3} = 11.28\,\text{mA} \approx 11.29\,\text{mA}.$$

Part B: Maximum Power Transfer Theorem

For $V_{th} = 8\,\text{V}$ and $R_i = 663\,\Omega$:

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad P_L = I_L^2 R_L.$$

Peak power occurs at $R_L = R_{th} = 663\,\Omega$:

$$P_{L(\max)} = \frac{V_{th}^2}{4R_{th}} = 24.13\,\text{mW} \approx 24.12\,\text{mW}.$$

Data Collection

Part A: Superposition Theorem

Table 2: Superposition Verification – Trial 1 ($V_1 = 10\,\text{V}$, $V_2 = 10\,\text{V}$)

Configuration	V_1 **(V)**	V_2 **(V)**	**Measured** I_{R_3} **(mA)**	**Calculated** I_{R_3} **(mA)**
Both Sources Active	10	10	8.61	8.59
Only V_1 Active	10	0	5.36	5.39
Only V_2 Active	0	10	3.21	3.19

Table 3: Superposition Verification – Trial 2 ($V_1 = 15\,\text{V}$, $V_2 = 10\,\text{V}$)

Configuration	V_1 **(V)**	V_2 **(V)**	**Measured** I_{R_3} **(mA)**	**Calculated** I_{R_3} **(mA)**
Both Sources Active	15	10	11.3	11.29
Only V_1 Active	15	0	8.11	8.09
Only V_2 Active	0	10	3.2	3.19

Part B: Maximum Power Transfer Theorem

Table 4: Power Transfer Measurement Log ($V_{th} = 8\,\text{V}$, $R_i = 663\,\Omega$)

S.No.	R_L **(Ω)**	**Measured** I_L **(mA)**	**Calculated** I_L **(mA)**	**Observed** P_L **(mW)**	**Calculated** P_L **(mW)**
1	99.6	10.70	10.50	11.30	10.92
2	327	8.22	8.08	22.10	21.80
3	460	7.23	7.12	24.00	23.30
4	**663 (Peak)**	**6.14**	**6.03**	**24.90**	**24.12**
5	973	4.97	4.89	23.90	23.20
6	2152	2.88	2.84	17.80	17.30
7	3232	2.07	2.05	13.90	13.60

Result Analysis

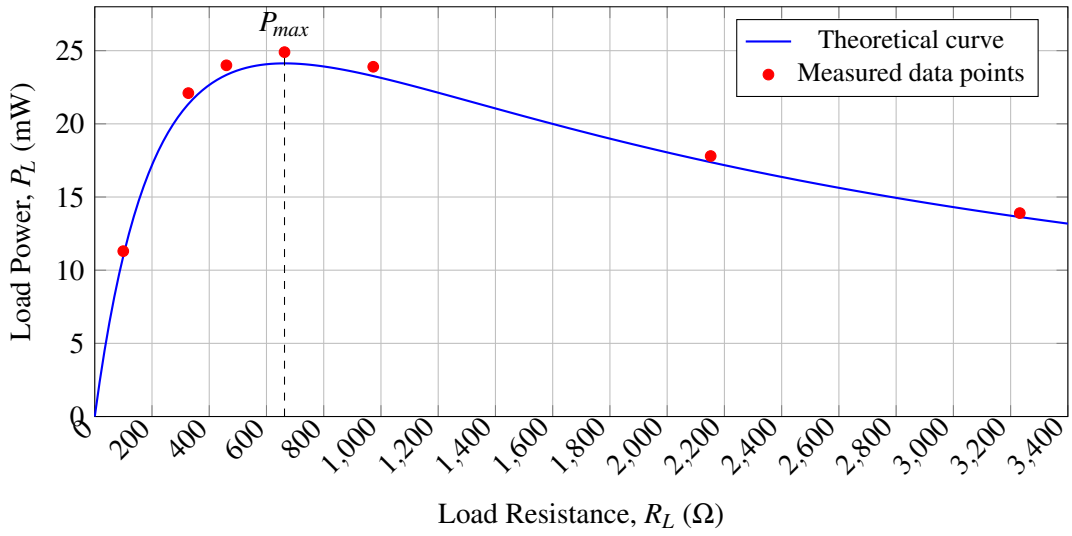


Figure 5: Load power P_L versus load resistance R_L, illustrating peak delivery at $R_L = R_i = 663\,\Omega$.

Part A: Superposition Theorem

The algebraic summation of individual current responses closely matches the multi-source total:

- **Trial 1**: $I'_{R_3} + I''_{R_3} = 5.36\,\text{mA} + 3.21\,\text{mA} = 8.57\,\text{mA}$ (Measured combined = 8.61 mA, percentage error $= 0.23\%$).
- **Trial 2**: $I'_{R_3} + I''_{R_3} = 8.11\,\text{mA} + 3.2\,\text{mA} = 11.31\,\text{mA}$ (Measured combined = 11.3 mA, percentage error $= 0.09\%$).

Part B: Maximum Power Transfer Theorem

Peak load power ($P_L = 24.90\,\text{mW}$) was recorded at $R_L = 663\,\Omega$, matching internal series resistance $R_i = 663\,\Omega$. The deviation between observed and calculated maximum power is 3.23%.

Discussion and Sources of Error

Experimental deviations remained under 3.5%, influenced by:

1. **Resistor Tolerances:** Discrepancies between nominal and real component values.
2. **Supply Regulation:** Minor fluctuations in supply output voltage under varying branch currents.
3. **Instrument Loading:** Non-ideal internal resistance of measurement devices.
4. **Decade Box Resolution:** Small contact and lead resistances within the variable resistance box.

Conclusion

Both network theorems were confirmed experimentally within minimal error bounds. Superposition held true for linear currents across independent source configurations. Furthermore, maximum power delivery occurred precisely when load resistance equaled source impedance.

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3. Boylestad, R. L. (2015). *Introductory Circuit Analysis* (13th ed.). Pearson.